COURSE PROJECT SYNOPSIS

1. Project Title

Autonomous Generative Lightweighting of SpaceBorne Mirrors

2. Team Member Details

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3. Problem Statement

Space-borne telescope mirrors must be extremely light yet optically stable under launch loads and the orbital thermal-gravitational environment. Conventional lightweighting depends on manual CAD pocketing and trial-and-error Finite Element Analysis (FEA), which takes weeks of design iteration. There is a need for an autonomous, physics-informed AI agent that generates multi-topology lightweight pocket structures (Isogrid, Honeycomb, Voronoi, Auxetic), predicts structural deformation in real time, and optimizes the mirror for over 75% mass reduction while keeping surface figure error $RMS_{sag} \leq \lambda/20$ and first natural frequency $f_1 \geq 120$ Hz.

4. Project Objectives

1) To develop a parametric generative CAD agent that synthesizes eight lightweight pocket topologies with an integrated 18-point whiffletree mount.
2) To train a physics-informed neural surrogate (PINN / GNN / MLP) that predicts mass, surface figure error, 37-term Zernike coefficients and f_1 in sub-second time.
3) To implement a reinforcement learning (PPO / SAC) and evolutionary optimizer that balances mass, optical figure and launch stiffness across Zerodur, SiC, ULE and beryllium.

5. Proposed Methodology

The project follows an end-to-end intelligent engineering pipeline:

1. **Parametric Modeling & Data Generation:** Design variables (aperture, faceplate and rib thickness, pocket span, material) are sampled. NASTRAN SOL 101 (static gravity sag) and SOL 103 (modal frequency) generate training data, supplemented by NASA and Mendeley datasets.
2. **Deep Learning Surrogate Model:** A multi-head neural surrogate replaces slow FEA solvers, mapping geometric and material inputs to 2D surface deformation maps and scalar metrics in milliseconds.
3. **Optimization Engine:** An RL policy coupled with a Genetic Algorithm refines pocket layouts, penalizing skin-quilting buckling (NASA CR-124075 criteria) and optical wavefront distortion.

4. **CAD Export & Validation:** The optimal design is exported as a parametric 3D CAD model (.STEP / .PRT) and validated against space telescope flight baselines (JWST, HST).

Proposed Workflow:

User Constraints (Aperture, Material, Launch Specs) → Generative Pocket Engine (8 Topologies) → Physics-Informed Surrogate (Mass / Sag / Frequency) → Multi-Objective RL/GA Optimizer (Buckling & Zernike Loss) → Automated 3D CAD Generation (.STEP / Siemens NX) → Verification against NASA & Flight Baselines

6. Datasets

Dataset	Source	Description	No. of Samples / Records	Purpose
Optical Mirror Surface Deformation & Zernike Expansion Dataset	Mendeley Data / Harbin Institute of Technology *(DOI: 10.17632/xpxhpdjgx9.1)*	Finite element mirror surface deformation matrices with 37-term Fringe Zernike polynomial coefficients.	**2,200 rows × 49 columns**	To train and test the surrogate to predict surface figure error, Zernike coefficients (Z_1–Z_{37}) and quilting patterns.
NASA CR-124075 Isogrid Structural & Buckling Testing Dataset	NASA Technical Reports Server (NTRS), *CR-124075*	Isogrid rib geometries, skin-to-rib area ratios (α), height ratios (δ), skin buckling thresholds, rib crippling limits and instability loads.	**3,884 records × 17 columns**	To set constraint envelopes in the RL reward function that prevent rib failure and skin quilting.

7. Algorithm / Technique Planned

Not Decided.

8. Evaluation Metrics

Not Planned.

9. Expected Results

Not Planned.